B.TECH DATA SCIENCE

Module 2: Statistics and Probability

B.Tech Data Science — Expanded Study Notes and Exam Preparation

Contents

Four chapters of expanded concepts, definitions, examples, applications, methods and revision questions.

Includes side headings on every study page and important questions at the end for exam preparation.

Prepared for learning and exam revision. Follow your university syllabus and faculty guidance where they differ.

1.1 Introduction to Statistics

Study focus: Statistics and Probability. Read the side headings, understand the example, and revise the key ideas.

Meaning

Statistics is the discipline of collecting, organising, summarising, analysing and interpreting data. It helps describe a dataset and reason about a larger population from observations.

Descriptive statistics

Descriptive statistics summarise the observations actually collected. Tables, graphs, mean, median, variance and standard deviation are common tools.

Inferential statistics

Inferential statistics uses sample data to estimate population quantities or test claims. Its conclusions depend on the sampling process, assumptions and uncertainty.

Population and sample

A population is the full group of interest; a sample is the subset observed. Studying every member may be costly, so samples are often used. A biased sample can lead to misleading conclusions.

Applications

Statistics supports quality control, healthcare studies, marketing, forecasting, education research, machine learning evaluation and public policy. Sound interpretation is as important as calculation.

1.2 Types of Variables and Data

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Categorical variables

Categorical variables represent groups. Nominal variables have no natural order, such as blood group or product category. Ordinal variables have a meaningful order, such as satisfaction levels.

Numerical variables

Discrete variables are countable, such as number of defects. Continuous variables can take values across an interval, such as weight or processing time.

Measurement scales

Nominal, ordinal, interval and ratio scales differ in the operations that make sense. A ratio variable has a meaningful zero, while an interval variable may not.

Frequency tables

A frequency table records how often each value or category occurs. Relative frequency divides a category count by the total; percentage frequency multiplies the relative frequency by 100.

Choosing a summary

The type of variable and the shape of its distribution guide the summary. Category counts suit categorical variables, while numerical variables may need measures of centre and spread.

1.3 Mean, Median and Mode

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Arithmetic mean

The mean is the sum of observations divided by the number of observations: mean = Σx/n. It uses every value and is useful for balanced numerical data.

Median

The median is the middle observation after sorting. For an odd number of observations, it is the central value; for an even number, it is commonly the average of the two central values.

Mode

The mode is the most frequent value or category. A dataset may have one mode, multiple modes or no uniquely most frequent value.

Outlier sensitivity

The mean is affected by extreme observations, while the median is often more resistant. For income data with a few very large values, the median may better represent a typical observation.

Worked example

For 2, 3, 3, 7, 10, the mean is 25/5 = 5, the median is 3 and the mode is 3. These statistics answer different questions about the same data.

1.4 Range, Variance and Standard Deviation

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Range

Range = maximum – minimum. It is simple to calculate but depends only on the two extreme observations.

Variance

Variance measures average squared deviation around the mean. Population variance divides the sum of squared deviations by N; sample variance commonly divides by n–1 to estimate population variance.

Standard deviation

Standard deviation is the square root of variance. It is expressed in the original unit and describes typical spread around the mean, though interpretation depends on distribution shape.

Interpretation

A small standard deviation indicates values are relatively close to the mean; a large one indicates more spread. It should not be interpreted without considering the data scale and context.

Example

For values 2, 4, 6, the mean is 4 and deviations are –2, 0, 2. Squared deviations sum to 8. Population variance is 8/3; population standard deviation is sqrt(8/3).

1.5 Quartiles, IQR and Outliers

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Quartiles

Quartiles divide ordered observations into portions. Q1 represents the lower quarter boundary, Q2 is the median, and Q3 represents the upper quarter boundary. Exact calculation conventions can vary.

Interquartile range

IQR = Q3 – Q1. It describes the spread of the middle half of observations and is less sensitive to extreme values than the range.

Box plot

A box plot commonly displays Q1, median and Q3, with whiskers extending according to a chosen convention. Points beyond the whiskers may be flagged as potential outliers.

Potential outliers

A common screening rule flags values below Q1 – 1.5×IQR or above Q3 + 1.5×IQR. This is a rule for investigation, not proof that the values are errors.

Good practice

Check the source, units and collection process before changing an extreme value. Rare but genuine events may be the most important cases to understand.

2.1 Basic Probability

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Definition

Probability quantifies uncertainty on a scale from 0 to 1. A probability of 0 indicates an impossible event under the model, and 1 indicates a certain event.

Sample space and events

The sample space contains all possible outcomes of an experiment. An event is a set of outcomes, such as rolling an even number on a fair six-sided die.

Equally likely outcomes

When outcomes are equally likely, P(A) = favourable outcomes / total outcomes. This shortcut does not apply when outcomes have unequal probabilities.

Complement rule

P(not A) = 1 – P(A). If the probability of rain is 0.3 under a stated forecast model, the probability of no rain is 0.7 for the same event definition.

Probability models

A probability model must assign valid probabilities and account for the possible outcomes. Assumptions such as fairness or independence should be stated rather than silently presumed.

2.2 Addition and Multiplication Rules

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Addition rule

For events A and B, P(A or B) = P(A) + P(B) – P(A and B). The overlap is subtracted because it would otherwise be counted twice.

Mutually exclusive events

If A and B cannot occur together, P(A and B) = 0. The addition rule simplifies to P(A or B) = P(A) + P(B).

Multiplication rule

P(A and B) = P(A)P(B|A), where P(B|A) is the probability of B given A. If A and B are independent, this simplifies to P(A)P(B).

Independence

Independence means learning that one event occurred does not change the probability of the other under the model. Mutually exclusive events with nonzero probabilities are not independent.

Example

For a fair coin and a fair die, the probability of heads and rolling a 6 is 1/2 × 1/6 = 1/12 if the two experiments are independent.

2.3 Conditional Probability and Bayes' Rule

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Conditional probability

P(A|B) = P(A and B)/P(B), provided P(B) > 0. It measures the probability of A when B is known to have occurred.

Reading a table

For conditional probability, identify the conditioning group first. The denominator is the total probability or count for that group, not necessarily the entire dataset.

Bayes' rule

Bayes' rule reverses conditional probabilities: P(A|B) = P(B|A)P(A)/P(B). It combines prior information with how likely the evidence is under each possibility.

Practical example

A positive screening result does not automatically mean a person has a condition. Interpretation depends on prevalence, sensitivity and false-positive rate, so base rates matter.

Common errors

Do not confuse P(A|B) with P(B|A). State the event and denominator clearly, and check that conditional probabilities are between 0 and 1.

3.1 Random Variables and Distributions

Study focus: Statistics and Probability. Read the side headings, understand the example, and revise the key ideas.

Random variable

A random variable assigns a numerical value to each outcome of a random experiment. Discrete variables have countable possible values; continuous variables take values over intervals.

Probability mass function

For a discrete random variable, a probability mass function gives the probability of each possible value. Probabilities are non-negative and sum to 1.

Probability density

For a continuous variable, probabilities are represented by area under a density curve. The probability of an exact single point is generally zero in a continuous model.

Expected value

Expected value is the probability-weighted long-run average. For a discrete variable, $E(X) = \Sigma\, xP(X=x)$. It need not be a value that occurs in a particular observation.

Variance of a random variable

Variance describes spread around the expected value and can be expressed as $E[(X–E(X))^2]$. Standard deviation is the square root of variance.

3.2 Bernoulli and Binomial Distributions

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Bernoulli distribution

A Bernoulli trial has two outcomes, often labelled success and failure, with success probability p. The random variable is 1 for success and 0 for failure.

Binomial conditions

A binomial model counts successes in a fixed number n of independent trials, each with the same success probability p. The conditions should be checked before using the formula.

Probability formula

For X ~ Binomial(n,p), P(X=k) = C(n,k)p^k(1–p)^(n–k), for k from 0 to n.

Mean and variance

The binomial mean is np and its variance is np(1–p). These describe the expected count and its spread across repeated sets of trials.

Example and caution

If a fair coin is tossed 4 times, the number of heads follows Binomial(4, 0.5) under the usual independence assumption. Trials with changing probabilities may require another model.

3.3 Normal Distribution and Empirical Rule

Study focus: Statistics and Probability. Read the side headings, understand the example, and revise the key ideas.

Shape

The normal distribution is a symmetric, bell-shaped continuous distribution described by mean μ and standard deviation σ. The mean, median and mode coincide in an ideal normal distribution.

Standardisation

The z-score $z = (x–\mu)/\sigma$ expresses a value in standard-deviation units relative to the mean. It helps compare values measured on different scales when the interpretation is appropriate.

Empirical rule

For an approximately normal distribution, about 68% of observations lie within one standard deviation of the mean, 95% within two, and 99.7% within three.

Uses

Normal models are used in measurement error, approximate sampling distributions and some natural measurements. Not every real-world dataset is normal, so shape should be examined.

Caution

A bell-shaped assumption should not be made merely because a calculation is convenient. Skew, outliers, truncation and mixtures can produce a very different distribution.

3.4 Sampling Distributions and the Central Limit Idea

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Sampling distribution

A sampling distribution describes how a statistic, such as a sample mean, varies across repeated samples drawn under the same process.

Standard error

Standard error measures the variability of an estimator across samples. It is different from the standard deviation of individual observations.

Central limit idea

Under suitable conditions, the distribution of a sample mean becomes approximately normal as sample size grows, even when the original population is not normal. The conditions and dependence structure matter.

Sample size

Larger samples often reduce random sampling variability, but do not automatically eliminate bias, poor measurement or non-representative sampling.

Practical meaning

Inference quantifies uncertainty from sampling. It does not repair an invalid sample design or a poorly defined population.

4.1 Estimation and Confidence Intervals

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Point estimate

A point estimate is one numerical estimate of a population quantity, such as a sample mean used to estimate a population mean.

Interval estimate

A confidence interval gives a range produced by a procedure designed to achieve a stated long-run coverage rate under its assumptions.

Interpretation

A 95% confidence procedure would cover the true parameter in about 95% of repeated samples when assumptions hold. It does not mean there is a 95% probability that a fixed parameter lies in this already computed interval under the frequentist interpretation.

Width and precision

Intervals tend to be narrower with more information and wider when variability is high. Narrowness is not enough: the method must also be appropriate.

Reporting

Report the estimate, interval, units, sample size and method. Explain practical importance as well as statistical uncertainty.

4.2 Hypothesis Testing and p-values

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Hypotheses

The null hypothesis H0 represents a reference claim; the alternative hypothesis H1 represents a competing claim. The test is designed before examining results where possible.

Test statistic

A test statistic summarises how far the observed result is from what the null model predicts, taking variability into account.

p-value

A p-value is the probability, assuming H0 and the model assumptions are true, of obtaining a result at least as extreme as the observed result according to the test definition.

Decision and significance

A p-value may be compared with a preselected significance level, but it does not measure effect size or practical value. Failing to reject H0 does not prove H0 is true.

Common mistakes

The p-value is not the probability that H0 is true, nor the probability that the result happened by chance in a general sense. Report estimates and uncertainty alongside it.

4.3 Correlation, Causation and Confounding

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Correlation

Correlation measures the direction and strength of association between variables. Pearson correlation captures linear association; other measures may be appropriate for ranks or nonlinear relationships.

Causation

Causation means that changing one factor would change an outcome under a defined causal interpretation. Observational association alone usually cannot establish this.

Confounding

A confounder is related to both the factor of interest and the outcome and can help explain an observed association. For example, season may affect both ice-cream sales and swimming incidents.

Study design

Randomised experiments can support causal conclusions when implemented well. Observational studies may use careful design and adjustment, but unmeasured confounding can remain.

Responsible interpretation

Use cautious language, consider alternative explanations, examine study design and avoid claiming causation from a scatter plot or correlation coefficient alone.

5. Important Exam Questions and Revision

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Numerical problems

1. Calculate mean, median, mode and range for a dataset. 2. Calculate variance and standard deviation. 3. Solve an addition-rule problem. 4. Find a conditional probability from a table. 5. Solve a binomial probability question.

Long-answer questions

1. Explain descriptive and inferential statistics. 2. Discuss probability addition and multiplication rules. 3. Explain the normal distribution and empirical rule. 4. Describe hypothesis testing and p-values. 5. Differentiate correlation and causation.

Short-answer questions

1. Define population and sample. 2. What is IQR? 3. Define independent events. 4. What is a random variable? 5. Define standard error. 6. What is a confidence interval?

Formula revision

Mean = Σx/n. Range = maximum – minimum. IQR = Q3 – Q1. P(not A)=1–P(A). P(A or B)=P(A)+P(B)–P(A and B). P(A|B)=P(A and B)/P(B). Binomial: P(X=k)=C(n,k)p^k(1–p)^(n–k).

Exam tip

Write the formula, substitute the given values, show intermediate steps and state the final answer with units or interpretation. In theory answers, explain assumptions and give a practical example.